# Predicting F1 Lap Time

Machine learning on FastF1 race data, 2022 to 2025. All models implemented from scratch with NumPy and pandas.


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## 1. PROBLEM

Teams plan pit stops and race pace by forecasting lap times. We predict how many seconds slower or faster a lap will be than the race's field-median lap, from the tyre and its wear, the fuel load, the weather, the car's recent form and the traffic.

## 2. DATA

Lap-by-lap timing, tyre and weather data from the FastF1 API for 92 races from 2022 to 2025. We keep dry, green-flag laps only: lap 1, pit laps, safety car, virtual safety car and yellow-flag laps, rain, deleted laps and laps slower than 104% of the driver's best are removed. That leaves 70,830 laps from 90 races. Fuel burn (about 2.95 s over a full race) is estimated from the data.

## 3. METHOD

**Target.** Lap time minus the race's field-median lap time, in seconds. This removes circuit length and year-to-year pace differences.

**Features.** Tyre compound and age (with non-linear terms and an age-by-compound interaction), lap number and race fraction as a fuel proxy, stint, track and air temperature, humidity, wind, driver and team form (their average pace over earlier races only), traffic (gaps to the cars ahead and behind and position at the end of the previous lap), field context (the field's pace trend and laps since the last safety car or yellow flag), and the circuit's elevation change per kilometre.

**Model.** Gradient boosting with regression trees, written from scratch (Friedman's LAD-TreeBoost generalised to a pinball, or quantile, loss). Features are binned into histograms; each round fits a tree to the negative gradient of the loss, with row and column subsampling, and each leaf is set to the quantile of the residuals it contains. At the median this is the absolute-error loss, which is robust to traffic and mistakes. Quantile versions at 5% and 95%, calibrated with split conformal prediction, give 90% prediction intervals. Ridge regression and least squares are also implemented in NumPy. Against XGBoost on the same split the implementation differs by under 0.002 s in MAE.

**Three tiers.** The pre-lap model uses everything known at the start of the lap except the driver's own previous lap times. The pre-stint model adds how the driver went in the earlier stints of the same race (their mean pace, the pace at the end of the previous stint and its length). The live model also adds the driver's previous clean laps in the same stint, like a pit-wall forecast.

**Baselines.** The median lap; persistence (the driver's last 3 laps averaged); Formula A, the structure of an online tyre calculator (base plus compound, track temperature, air temperature and abrasiveness factors times laps) with constants fitted to our data; Formula B, Formula A plus fuel and form; and ridge regression on the same features as the pre-lap model.

**Validation.** Training on 2022 to 2024 (52,092 laps, 66 races) and testing on all of 2025 (18,738 laps, 24 unseen races). Mixed-year 5-fold cross-validation by race, where each fold holds out about 20% of every year's races, and leave-one-year-out. Laps from one race are never split between training and test.

## 4. RESULTS

Unseen 2025 races. MAE is the typical error in seconds; curve error compares predicted and actual average lap-time loss by tyre age.

| Model | MAE (s) | Median AE (s) | $R^2$ | Curve error (s) | Slope corr. |
|---|---|---|---|---|---|
| Median baseline | 0.875 | 0.765 | 0.00 | 0.203 | -0.29 |
| Formula A (calculator style) | 0.862 | 0.753 | 0.02 | 0.124 | -0.09 |
| Formula B (fuel and form) | 0.588 | 0.453 | 0.46 | 0.083 | 0.28 |
| Ridge regression, pre-lap features | 0.503 | 0.393 | 0.60 | 0.083 | 0.33 |
| **Gradient boosting, pre-lap** | **0.468** | **0.367** | **0.66** | **0.063** | **0.57** |
| **Gradient boosting, pre-stint** | **0.437** | **0.342** | **0.70** | **0.067** | **0.61** |
| Persistence (last 3 laps) | 0.368 | 0.229 | 0.72 | 0.058 | 0.86 |
| **Gradient boosting, live** | **0.294** | **0.197** | **0.82** | **0.037** | **0.94** |

Persistence is the fair baseline for the live model. The live model improves on it by 20% in MAE.

Mixed-year cross-validation across all 90 races, the main performance estimate:

| Model | MAE (s) | $R^2$ | Fold MAE std |
|---|---|---|---|
| Median baseline | 0.898 | 0.00 | 0.032 |
| Formula A (calculator style) | 0.887 | 0.02 | 0.030 |
| Formula B (fuel and form) | 0.571 | 0.53 | 0.027 |
| Ridge regression, pre-lap features | 0.520 | 0.61 | 0.029 |
| **Gradient boosting, pre-lap** | **0.494** | **0.64** | **0.032** |
| **Gradient boosting, pre-stint** | **0.461** | **0.68** | **0.028** |
| Persistence (last 3 laps) | 0.387 | 0.71 | 0.012 |
| **Gradient boosting, live** | **0.310** | **0.81** | **0.013** |

Laps in stint 2 or later (70% of 2025 laps), where the pre-stint model can use the driver's earlier stints:

| Model | MAE (s) | Median AE (s) | $R^2$ |
|---|---|---|---|
| Median baseline | 0.859 | 0.730 | -0.10 |
| Persistence (last 3 laps) | 0.384 | 0.241 | 0.66 |
| **Gradient boosting, pre-lap** | **0.500** | **0.392** | **0.56** |
| **Gradient boosting, pre-stint** | **0.457** | **0.356** | **0.63** |
| **Gradient boosting, live** | **0.315** | **0.209** | **0.78** |

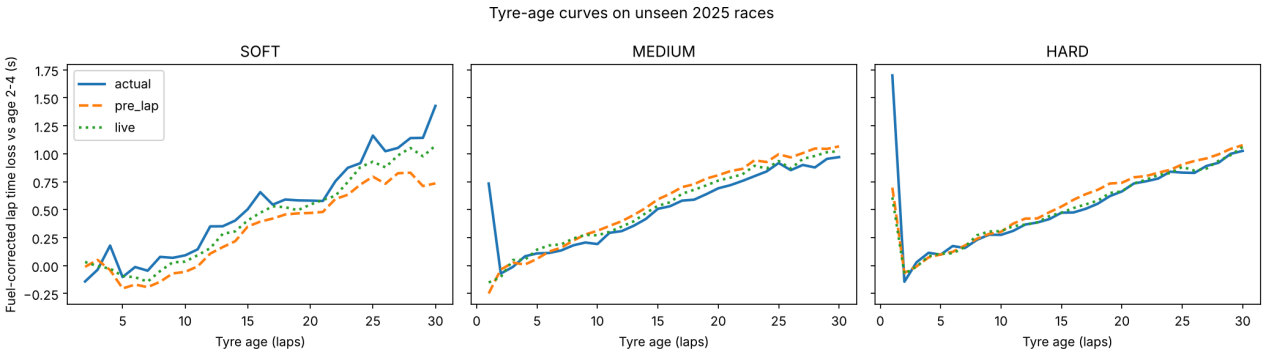



Figure 1. Fuel-corrected lap-time loss by tyre age on unseen 2025 races: actual and predicted.

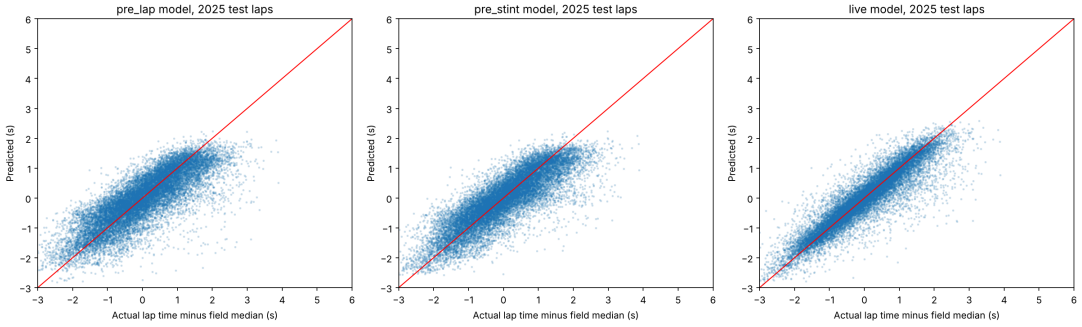



Figure 2. Predicted against actual lap time relative to the field median, 2025 test laps.

## 5. THE APP

A Streamlit app with two tabs makes the models explorable. **Lap times** runs the from-scratch gradient boosting models (pre-lap and live) on laps the user types, pastes or uploads (lap number, lap time, compound, tyre life, track temperature and an optional circuit) and shows predicted against actual lap times; inputs left out take neutral defaults. **Model** shows the scores from this report and a button that re-runs the saved models on all 2025 laps they never saw and computes MAE and $R^2$ on the spot.

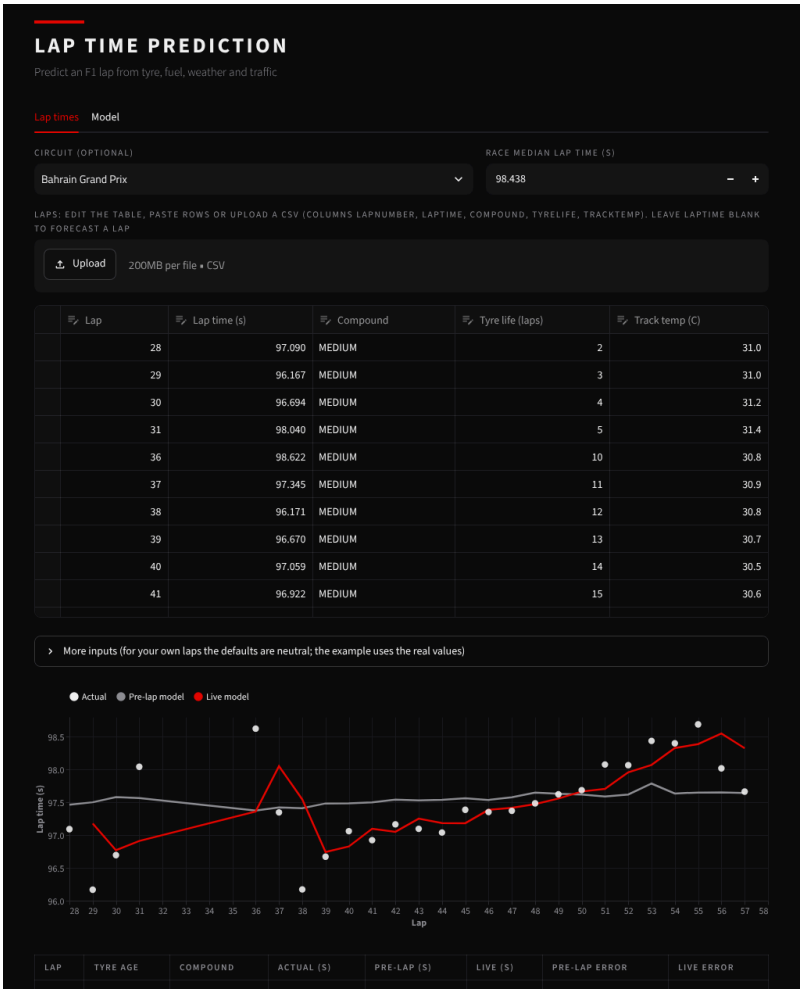



Figure 3. Lap times tab: the pre-lap and live gradient boosting models against the real laps of one stint (the example stint is part of the training data).

## 6. VALIDATION CHECKS

- **No leakage.** A model trained on shuffled targets scores $R^2$ 0.00 and MAE 0.87, the same as the baseline.
- **Not a lucky season.** Leave-one-year-out gives similar errors for every year (below), so regulation changes do not break the model.
- **Overfitting.** Pre-lap train MAE 0.374 against test 0.468; live 0.261 against 0.294.
- **Not by chance.** Resampling the 24 test races 1,000 times, the pre-lap MAE has a 95% interval of 0.43 to 0.52 s and beats Formula B by 0.12 s (0.08 to 0.16) in every resample.
- **Intervals.** The calibrated 90% prediction intervals cover 93% (pre-lap) and 93% (pre-stint) and 93% (live) of 2025 laps, with mean widths of 2.2 s, 2.1 s and 1.5 s.

Leave-one-year-out, MAE in seconds ($R^2$):

| Held-out year | Pre-lap | Pre-stint | Live |
|---|---|---|---|
| 2022 | 0.537 (0.62) | 0.499 (0.67) | 0.336 (0.81) |
| 2023 | 0.471 (0.64) | 0.442 (0.67) | 0.301 (0.81) |
| 2024 | 0.477 (0.67) | 0.458 (0.69) | 0.313 (0.82) |
| 2025 | 0.468 (0.66) | 0.437 (0.70) | 0.294 (0.82) |

Feature-group ablation, pre-lap model, 2025 MAE in seconds:

| Features | MAE (s) |
|---|---|
| **All features** | **0.468** |
| Without tyre | 0.512 |
| Without fuel progress | 0.510 |
| Without weather | 0.469 |
| Without form | 0.472 |
| Without traffic | 0.517 |
| Without field context | 0.476 |
| Without circuit | 0.465 |

Removing the tyre features costs about 0.04 s, about as much as removing fuel progress or traffic (0.05 s): tyre age is one contributor among several.

## 7. LIMITATIONS

- A single lap carries about 0.33 s of random noise from traffic and mistakes, so lap-level $R^2$ has a ceiling near 0.9 for any model; most pre-lap error is each driver's race-specific offset, which no pre-lap information reveals.
- The only circuit information is elevation change per kilometre, measured once per circuit from one fast lap; it changes the pre-lap error by only about 0.003 s.
- The target is relative to the race's field-median lap, which comes from the session itself, not from a separate forecast.
- The live model is mostly persistence: the plain last-3-lap average already reaches $R^2$ 0.72.
- Some design choices (the lap filter and feature set) were made while looking at 2025 results, so the 2025 holdout is slightly optimistic; cross-validation is the better estimate.

## 8. CONCLUSION

A from-scratch gradient boosting model predicts lap time relative to the field with MAE 0.468 s ($R^2$ 0.66) using only information available before the lap, against 0.875 s for a naive baseline and 0.588 s for a fitted formula, and reproduces real tyre-age curves on unseen races. Knowing how the driver went in earlier stints cuts the error to 0.437 s ($R^2$ 0.70). With the driver's recent laps, error drops to 0.294 s, 20% better than simple persistence.